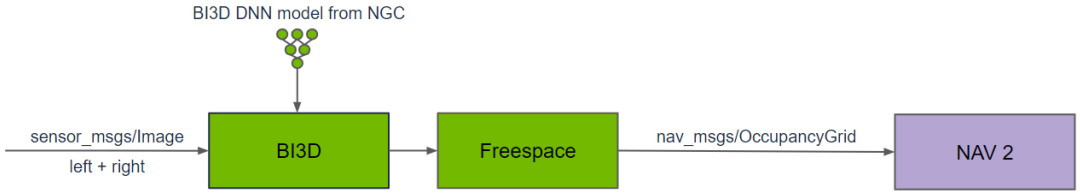
Isaac ROS Free Space Segmentation

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- Overview
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Isaac ROS Free Space Segmentation official website link: https://nvidia-isaac-ros.github.io/repositories_and_packages/isaac_ros_freespace_segmentation/index.html

Overview

Free space segmentation can process segmentation masks that contain the pose of the robot relative to the ground, and generate occupancy grids for obstacle avoidance for Nav2. Bi3D free space is an additional source of occupancy grids that mobile robots can use.



Quick Experience

First use the command below to start the container

```
./run_isaac.sh
```

Then confirm the container on the host with `docker ps`, and open container terminals with `docker exec -it <container name> bash`; each terminal should use the same ROS/FastDDS environment, for example:

```
Jetson Orin Nano L4T R39.2.1 · (8GB + 8GB ZRAM)
RDP 192.168.2.61:3389 jetson/yahboom [LAN]
VNC 192.168.2.61:5900 pass: yahboom [LAN]
jetson@jetson:~$ docker ps
CONTAINER ID   IMAGE                          COMMAND                  CREATED
      STATUS          PORTS     NAMES
8f397e4b6c3e   my-isaac-ros-slam:4.6-all   "/bin/bash -c '\ncat …"   42 seconds
ago   Up 41 seconds             magical_bhabha
jetson@jetson:~$
```

Note: The terminals below all execute inside the container:

> After the power mode is set to MAXN, an overcurrent warning may pop up when running the case; ignore it.

Terminal A: Start Nano pool=8 ESS. The input image is 1920×1080, and the model input is 960×576:

```
ros2 launch /workspaces/isaac_ros-dev/launch/ess_nano_pool8.launch.py \
  engine_file_path:=/workspaces/isaac_ros-
dev/isaac_ros_assets/models_pending/dnn_stereo_disparity/dnn_stereo_disparity_v4.
1.0_onnx_trt10.16/ess.engine \
  ess_plugin_file_path:=/workspaces/isaac_ros-
dev/isaac_ros_assets/models_pending/dnn_stereo_disparity/dnn_stereo_disparity_v4.
1.0_onnx_trt10.16/plugins/aarch64/ess_plugins.so \
  input_image_width:=1920 input_image_height:=1080 \
  model_input_width:=960 model_input_height:=576
```

Terminal B: Play the input in a loop at 0.25× speed:

```
ros2 bag play /workspaces/isaac_ros-
dev/isaac_ros_assets/isaac_ros_ess/rosbags/ess_rosbag \
  --loop --rate 0.25 --remap \
  /left/camera_info:=/left/camera_info_rect \
  /right/camera_info:=/right/camera_info_rect
```

Terminal C: Run the depth segmentation post-processing:

```
python3 /workspaces/isaac_ros-dev/ess_segmentation.py
```

Terminal D: Open rqt image view and select `/ess/depth_segmentation` in Topic:

```
ros2 run rqt_image_view rqt_image_view
```



